

LAB ASSIGNMENT 19.1

Task 1 – Python to Java Conversion

Task:

Translate a simple **Python class** into **Java** using AI assistance.

Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

Python code :

```
week 19 > class.py > ...
1  #write a python program with a class and attributes and methods
2  class Person:
3      def __init__(self, name, age):
4          self.name = name
5          self.age = age
6
7      def greet(self):
8          print(f"Hello, my name is {self.name} and I am {self.age} years old.")
9  # create an instance of the Person class
10 person1 = Person("Alice", 30)
11 # call the greet method
```

Output:

```
SyntaxError: Invalid Syntax
PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/User
neDrive/Desktop/AIAC/week 19/class.py"
Hello, my name is Alice and I am 30 years old.
```

Java Code :

```
week 19 > class.py > ...
14 #convert the above code into java
15 class Person {
16     private String name;
17     private int age;
18
19     public Person(String name, int age) {
20         this.name = name;
21         this.age = age;
22     }
23
24     public void greet() {
25         System.out.println("Hello, my name is " + name + " and I am " + age + " years old.");
26     }
27
28     public static void main(String[] args) {
29         Person person1 = new Person("Alice", 30);
30         person1.greet();
31     }
32 }
```

Output:

```
Output
Hello, my name is Alice and I am 30 years old.
=== Code Execution Successful ===
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ **function** for factorial calculation into Python.

Instructions:

- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

C++ Code :

```

week 19 > factorial.py > ...
1  #write a c ++ program to calculate the factorial of a number
2  #include <iostream>
3  using namespace std;
4
5  // Factorial function
6  int factorial(int n) {
7      if (n == 0)
8          return 1;
9      return n * factorial(n - 1);
10 }
11
12 int main() {
13     int n = 5;
14
15     cout << factorial(n);
16
17     return 0;
18 }

```

Output :

```

Output
120
=== Code Execution Successful ===

```

Python code :

```

21  #convert the above code into python
22  def factorial(n):
23      if n == 0:
24          return 1
25      return n * factorial(n - 1)
26  n = 5
27  print(factorial(n))

```

Output :

```

● PS C:\Users\User\OneDrive\Desktop\AIAC> & C:\Users\User\OneDrive\Desktop\AIAC\week 19\factorial.py"
120
○ PS C:\Users\User\OneDrive\Desktop\AIAC>

```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

SQL :

```
class.py  factorial.py  demo.py 9+  = demo.sql X
week 19 > demo.sql
1  SELECT name, salary FROM employees WHERE salary > 50000;
```

Output :

```
Output
name  salary
1    B   60000
2    C   70000

=== Code Execution Successful ===
```

Pandas Code ;

```
3  -- generate pandas code for the above SQL query
4  import pandas as pd
5  # Assuming you have a DataFrame named 'employees' with columns 'name' and 'salary'
6  result = employees[employees['salary'] > 50000][['name', 'salary']]
7  print(result)
8
```

Output :

```
Output
name  salary
1    B   60000
2    C   70000

=== Code Execution Successful ===
```

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will choose a **real-time use case** where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into **JavaScript** for browser automation.
- Translating a **Java method** into **C#** for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least **two different programming languages**.

Python code:

```
week 19 > multi.py > ...
1  # Add two numbers
2  a = 10
3  b = 20
4  print("Sum:", a + b)
5
```

Output:

```
PS C:\Users\User\OneDrive\Desktop\IAC\week 19\multi.py"
Sum: 30
```

Java Code:

```
7  #convert the above code into java
8  public class Main {
9      public static void main(String[] args) {
10         int a = 10;
11         int b = 20;
12         System.out.println("Sum: " + (a + b));
13     }
14 }
```

```
IAC/week 19/multi.py"
Sum: 30
```

